Mechanisms of Action Prediction

A Multi-Model Ensemble for Highly Imbalanced Multi-Label Classification

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Project Work in Machine Learning and Data Mining

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Kaggle LISH MoA Prediction Challenge

Focus on the Problem

Prediction task

Given the gene-expression and cell-viability profiles of human cells exposed to a chemical compound, predict which biological mechanisms of action (MoAs) are triggered.

- **206 binary targets**: each target represents one possible MoA.
- **Multi-label setting**: a sample may have no positive label, one label, or multiple labels.
- **Evaluation metric**: mean binary log loss across samples and targets.
- **Key challenge**: very sparse positives, strong target imbalance, and repeated measurements from the same compound.

Central methodological question

How can validation estimate generalisation to compounds that may not have appeared during training?

Problem Data and Objectives

Dataset

- 23,814 training rows:
  - 21,948 treated
  - 1,866 control (7.8%)
- 772 gene-expression features (g-*).
- 100 cell-viability features (c-*).
- `cp_time`: 24, 48, 72 h; `cp_dose`: D1, D2.
- 3,288 distinct compounds among treated rows.

Label distribution

- Positive rate: **0.373%** of treated-row/target cells.
- 34.2% of treated rows have no label.
- 57.1% have one; 8.7% have two or more.
- Positive examples per target: 1–832 (median 38).
- 53 targets have fewer than 20 positives.

Project objectives

- Build a reproducible cross-validated pipeline for imbalanced multi-label learning.
- Compare neural and gradient-boosted models; reduce variance with seed bagging.
- Design compound-aware validation and quantify row-wise optimism.
- Tune models without selecting on the outer validation fold; assess an ensemble.

Controls have no MoA by definition: excluded from training and assigned all-zero predictions at inference.

Notebooks Produced

Notebook	Purpose / output
`moa-training.ipynb`	Main training, cross-validation, nested blending, per-target and CV diagnostics; saves model artifacts and OOF predictions.
`moa-inference.ipynb`	Loads saved models and fold pipelines; generates `submission.csv` without retraining.
`moa-tuning.ipynb`	Tuning v1: single-stage Optuna search on row-wise folds.
`moa-tuning-row.ipynb`	Tuning v2: two-stage row-wise search and confirmation.
`moa-tuning-hybrid.ipynb`	Tuning v3: compound-aware search and multi-seed confirmation.
`moa-fold-diagnostics.ipynb`	Fold sizes, compound effects, target coverage, and protocol-gap analysis.
`moa-blend-offline.ipynb`	Offline recombination and comparison of OOF predictions.
`moa-test-drug-overlap-nn.ipynb`	Indirect analysis of train/test compound overlap.
`moa-knn-override-freq-drugs.ipynb`	Tests a nearest-neighbour override for frequent compounds (rejected).

Deliverable structure

Training and inference are separated. The training notebook produces the artifacts; inference only loads them and writes the submission.

Cross-Validation Protocols

All protocols use 5-fold iterative multi-label stratification (`random_state=42`) on treated rows. The compound identifier comes from `train_drug.csv`; test rows do not have this identifier.

Protocol	Grouping unit	Main property
`row`	Individual rows	No compound grouping; 99.4% of validation rows have their compound in the training fold.
`grouped`	Entire compound	No compound overlap across folds; fold sizes are imbalanced (maximum deviation: 25.6%).
`hybrid`	Compounds, except frequent ones	Compounds with more than 18 rows are split row-wise; fold sizes stay within 1.4% of the mean. About 9.8% of validation rows share a compound with training.

Why hybrid?

It trades off compound separation and fold-size balance. The threshold of 18 rows was selected from diagnostics for one splitter seed; it remains a design choice, not a proven optimum.

The inner split used for neural-network early stopping follows the same grouping rule as the outer folds.

Hyperparameter Tuning

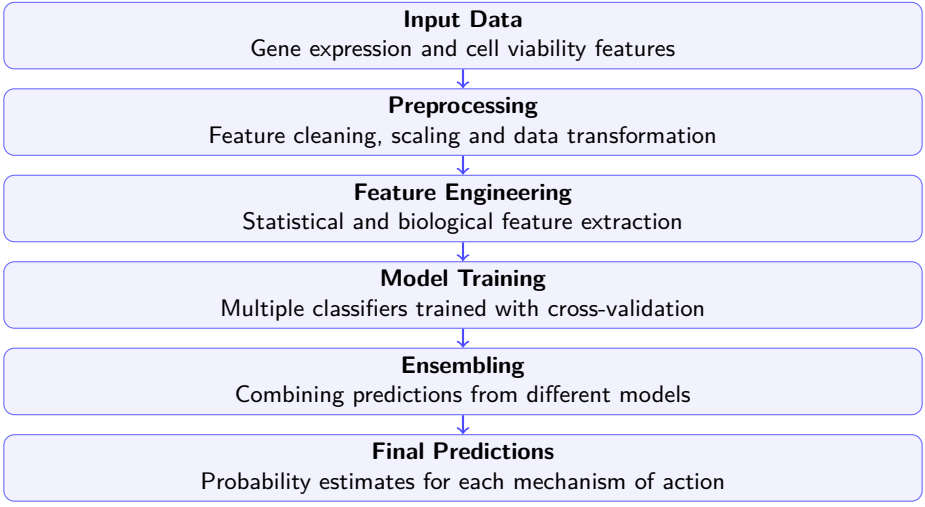
- Framework: **Optuna**, with TPE sampling and a median pruner.
- Three iterations of the tuning workflow:
  - v1: single-stage search on five row-wise folds; validation fold also selected the best epoch.
  - v2: search on row-wise fold 0, then confirm top candidates and defaults on fold 1.
  - v3: compound-aware search and confirmation, using three paired seeds.
- For v2/v3, a 10% inner split supports early stopping; the outer validation fold is scored after model selection.
- Adoption rules require a minimum gain over the default and confirmation on a second fold; v3 additionally requires a win on more than half of paired seeds.

Selected H2 settings

Model	Learning rate	Dropout	Other selected values
ResNet	$2.67 \cdot 10^{-3}$	0.583	768 hidden units; batch 512
FT-Transformer	10^{-3}	0.2	Defaults retained
CatBoost	0.1034	–	Depth 4; L2 = 56; 250 iterations

Tuning v3 did not find a better FT-Transformer configuration than the default. Several selected values lie near search-space boundaries.

End-to-End Pipeline



The pipeline combines preprocessing, feature engineering, model training and ensemble predictions.

Models Used

TabularResNet

- Residual neural network for tabular data.
- RankGauss + PCA preprocessing.
- 50 gene and 15 viability PCA components by default.
- Three-seed bagging.

FT-Transformer

- Attention-based tabular architecture.
- RankGauss; no PCA.
- 64 tokens, 2 blocks, 4 heads in reported runs.
- Three-seed bagging.

CatBoost

- Gradient-boosted decision trees.
- RankGauss; no PCA.
- MultiLogloss, 250 iterations.
- One model per fold.

Shared training choices

- Neural loss: `BCEWithLogitsLoss` with label smoothing 0.001.
- Optimiser: AdamW; ReduceLROnPlateau; mixed precision.
- Early stopping (patience 8), dropout, batch normalisation and weight decay.
- CatBoost targets with no positive training examples use a smoothed Jeffreys prior.

Preprocessing and Ensemble

Model-specific preprocessing

- **ResNet:** QuantileTransformer (RankGauss) → PCA separately on gene and cell-viability blocks → concatenation.
- **FT-Transformer / CatBoost:** QuantileTransformer (RankGauss), without PCA.

Ensemble strategy

- Average predictions across three seeds for each neural architecture.
- Combine ResNet, FT-Transformer and CatBoost using softmax-reparameterised weights.
- Optimise weights with Nelder–Mead on out-of-fold predictions.
- Re-validate weights through nested CV: fit on four folds and score on the fifth, rotating the held-out fold.

$$\hat{p}_{\text{ens}} = w_R \hat{p}_R + w_F \hat{p}_F + w_C \hat{p}_C, \qquad w_R + w_F + w_C = 1$$

The nested estimate limits optimism from fitting blend weights; it does not remove possible bias in the CV design or model-selection process.

Results: Cross-Validation

Mean OOF binary log loss on treated rows (lower is better).

Model	R: row/default	H1: hybrid/v2	H2: hybrid/v3
Baseline (target prior)	0.02220	0.02233	0.02233
TabularResNet	0.01632	0.01744	0.01735
FT-Transformer	0.01643	0.01736	0.01744
CatBoost	0.01699	0.01782	0.01783
Nested ensemble	**0.01607 ± 0.00017**	**0.01714 ± 0.00031**	**0.01715 ± 0.00031**

- Ensemble improves on the best single model by about 1.5% in R and 1.2% in H2.
- In H2, each model contributes: removing any one worsens the nested score in all five folds.
- Hybrid fold variability is higher, consistent with less homogeneous folds when compounds are kept together.

Important interpretation

The +6.7% loss from R to H2 is **not a controlled protocol comparison**: code revision, hyperparameters and fold assignments also changed.

Results: Tuning and Kaggle Test

Does tuning help?

- H1 ensemble: 0.01714.
- H2 ensemble: 0.01715.
- Difference: only $+10^{-5}$.
- No offline recombination reached the predefined gain threshold.

Compound-aware validation

Hybrid CV is less optimistic than row-wise CV and is closer to the private test score after approximate rescaling.

Late Kaggle submissions

	R	H2
Private	0.01634	0.01643
Public	0.01865	0.01873

- Private scores differ by $9 \cdot 10^{-5}$.
- Public scores differ by $8 \cdot 10^{-5}$.
- One submission per configuration: no reliable ranking between them.

The test set was not used to tune hyperparameters or choose configurations. CV is on treated rows; comparison with all test rows uses an approximate treated-share rescaling (0.910).

Additional Analyses and Negative Results

- **Test compound overlap:** nearest-neighbour analysis suggests at least five of eight analysed frequent compounds occur in the test set at roughly training frequency.
- **Unresolved general overlap:** the proxy has AUC 0.516 and cannot reliably distinguish ordinary seen from unseen compounds.
- **Nearest-neighbour override:** all four tested variants worsen hybrid OOF loss; the override was rejected.
- **Extra architectures:** adding an MLP or a 1D-CNN did not improve the ensemble beyond differences comparable to run noise.
- **Final model choice:** retain the simpler three-model ensemble (ResNet + FT-Transformer + CatBoost).

Takeaway

More components or compound-specific post-processing did not produce a robust measurable gain on the available validation evidence.

Limitations

1. **Unknown test compound IDs:** overlap can only be inferred indirectly; ordinary-compound overlap remains unassessed.
2. **Hybrid threshold is heuristic:** 18 rows was chosen from fold-balance diagnostics for one seed; threshold sensitivity of CV loss was not measured.
3. **Non-controlled protocol comparison:** row-wise and hybrid runs differ in code, hyperparameters and folds.
4. **Selection bias:** tuning/model choices reused folds overlapping reported OOF folds; no untouched hold-out was reserved. Full nested selection was computationally prohibitive.
5. **Tuning uncertainty:** searches were limited; some selected parameters lie at search-space boundaries. A controlled tuned-vs-default hybrid comparison is missing.
6. **Preprocessing detail:** unsupervised transforms include inner-validation features when fitted on the training fold; this may slightly affect early stopping.
7. **Test-score uncertainty:** only two late submissions, one per configuration; public/private subsets differ substantially.
8. **Reproducibility and inference:** GPU non-determinism exists, and inference requires the separately saved model-artifact dataset.

Conclusions

- The project delivers a reproducible training/inference workflow for a sparse, 206-target multi-label task.
- Compound-aware CV exposes a more difficult generalisation setting than row-wise splitting.
- A three-model ensemble consistently outperforms its individual components in the reported nested-CV experiments.
- Tuning did not measurably improve the hybrid ensemble; the two test submissions are statistically indistinguishable given the available evidence.

Main lesson

For repeated-compound biological measurements, validation design can matter at least as much as model architecture. The estimated performance must be interpreted together with the assumptions and limitations of the split.

References


Kaggle. *Mechanisms of Action (MoA) Prediction*, 2020. `https://www.kaggle.com/competitions/lish-moa`


Gorishniy, Y., Rubachev, I., Khrulkov, V., and Babenko, A. *Revisiting Deep Learning Models for Tabular Data*. NeurIPS, 2021.


Prokhorenkova, L. et al. *CatBoost: unbiased boosting with categorical features*. NeurIPS, 2018.


Sechidis, K., Tsoumakas, G., and Vlahavas, I. *On the Stratification of Multi-Label Data*. ECML PKDD, 2011.


Loshchilov, I. and Hutter, F. *Decoupled Weight Decay Regularization*. ICLR, 2019.


Akiba, T. et al. *Optuna: A Next-generation Hyperparameter Optimization Framework*. KDD, 2019.


Pedregosa, F. et al. *Scikit-learn: Machine Learning in Python*. JMLR, 2011.